Assignment 01

Terminology and Perception

1 Terminology I

Define the following terms in a few sentences and explain how they are related, and differentiate between them. In each case, also give specific examples:

a. Medium
   A: "Medium" is the Latin origin of the word "media" and means "something in the middle" or "intermediary". It is sometimes used to refer to a singular instance of media.
b. Media
   A: Media is a term encompassing any means to transport or distribute information. More technically, it is defined as channels or systems which store, transmit and distribute data, typically intended for human perception. This includes text, graphics, animations, images, speech, music, video, and more. Various representations of different types of data exist so that they can be stored and transported (typically in a digital format) and finally presented.
c. Perception
   A: Perception describes the stimuli that a human's sensory organs will "pick up" (i.e. see, hear, feel, smell, ...) when a piece of media is presented to them. Common examples include music, noise, voice, text, images, video.
d. Representation
   A: Representation describes how these stimuli are encoded internally, typically for digital processing, storage and transport. Examples for possible representations are texts being encoded as ASCII characters, audio samples being stored as pulse code modulation (PCM) or still images saved using JPEG encoding.
e. Presentation
   A: Presentation encompasses the input and output of media, specifically what serves to capture the initial sensory information before it passes through the representation stage and what presents it back to a human, bridging the gap from a representation back to a stimulus. Means of presentation for input include keyboards, cameras, microphones, gamepads; means of presentation for output include speakers and displays. Paper is another interesting example that could be both used for input and output.
f. Storage
   A: Storage refers to the device, tool or more generally "means" where media can be saved. Examples include paper, disk, tape, CD, DVD, BluRay, SD cards, ...,

each having different capacities, suitability to different types of media and longevity.

g. Transport
A: As media is often shared, means of transports are required to allow representations of it to reach others. This typically occurs via (fiber optic) cables, air or vacuum.

h. Discrete media
A: Discrete media refers to media that represents a "snapshot" in time, that can essentially be fully processed at a given moment since it lacks a temporal component; it is static. Text, still images, drawings or maps are examples of discrete media – the presentation of them does not change over time while still allowing them to be processed in completeness.

i. Continuous media
A: Continuous media is the contrast to discrete media, having a temporal component or being dynamic in nature. Its presentation changes over time, which is necessary for this kind of media to be experienced in completeness. Examples include speech, noise, music, video, animation

2 Terminology II

Explain the terms "linear medium" and "nonlinear medium" and assign computer game, television program, DVD, hypermedia, radio program, Google Maps to these terms.

From your point of view, what is "multimedia", what multimedia systems / programs do you use, what challenges do users face, what are the challenges on a technical level, and why do you find multimedia exciting?

A: Linear media offers no navigation or interaction to the viewer, while non-linear media does offer navigation and/or interaction.

Computer games, DVDs, hypermedia (connection of information nodes via hyperlinks, e.g. internet) and Google Maps are therefore examples for non-linear media.

A singular television or radio program would thus be an example for linear media, however, if multiple different programs were available to switch between (e.g. ORF Internetradio), this would constitute non-linear media.

Multimedia describes the combination of different forms of media into one cohesive form of media. For instance, a movie is a form of multimedia as it combines video, audio and text (subtitles) into one medium that is meant to be experienced only in this combination. Arguably, most applications that I use on a home computer are multimedia systems/programs, be it a web browser where a single web page consists of text, images, video, sound and animation, a visual operating system itself, video games or even classically a media player such as VLC.

Given all these common-day examples, I would argue that users typically do not even need to think about all these types of media that they are constantly exposed to and for the most part, the integration of multimedia works seamlessly. Some friction still arises due to unsupported file formats which may cause grievances when trying to use media assets obtained from one program in another context; a common issue was for instance the lackluster support of WebP images that operating systems boasted, which can definitely be frustrating; a personal pet peeve of mine is the longstanding refusal of Apple's Finder to preview OGG Vorbis/Opus audio files, which has only recently been rectified.

Another grievance arises from streaming media in slow network environments which can cause stutters or quality degradation in media playback or when in a real-time application (e.g. video calls) where even comparatively small latencies severely degrade user experience.

Both of these issues also correspond to technical counterparts where different situations have different requirements that need adaptation, yet a user expects everything to "just work" regardless the situation. Some encodings prioritise quality, but this may come with increased demands on bandwidth or processing power. In situations with low bandwidth, quality may have to be sacrificed to achieve acceptable latency. On the technical side, the right "tool" for the right job must be chosen and often the user should not have to deal with such complexities directly. There is no single perfect codec for every situation, and indeed there are often new emerging solutions that improve performance in some respects. This also explains why not all user software can support the same formats, it may sometimes simply not make sense, and sometimes it takes time to catch up to standards that are already well-established in other domains.

Making information accessible through multiple sensory channels is one of the key achievements of multimedia systems. While text terminals still have their niche efficiency for me in some regards, modern multimedia systems make many jobs easier with intuitive representations of information through e.g. picture and sound. However, the immersion that they enable is what I consider the most exciting part, personally. Multimedia has enabled us to experience rich worlds and narratives other humans have crafted in brand new ways. Whether through videos, music, or video games, they allow me not just to experience stories and voices of others but express myself in the same way creatively. To give an example dear to my heart: Game development is a hobby of mine that I pursue with several other student friends and when viewing it through this lens, it truly is magical.

3 Visual and auditory systems

Describe how "seeing" and "hearing" works and relate it to the technical implementation in photo cameras, for example.

A: Seeing involves the eyes, its optic nerves and parts of the brain through which electromagnetic waves within a certain spectrum are detected, processed and ultimately made perceptible to us. An incoming light wave is focused through the eye's lens, and directed onto the retina, where the image is formed. The pupil can dilate or constrict to control how much light passes through. On the retina, rods and cones act as photosensitive receptors. There are many more rods (~113-4 million) than cones (~6-7 million); rods only exist in one type and allow perceiving differences in brightness or contrast and respond to low energy. Cones on the other hand exist in three types that have different sensitivities to different wavelengths/frequencies of light – in humans they exist in three types, L (long wavelength – red), M (medium wavelength – green) and S (short wavelength – blue). This allows humans to distinguish between different colours. The resulting signal is then converted into electro-chemical impulses and passed through the optic nerve into a specialised area of the brain where the signal is interpreted and processed, turning it into the actual image that we perceive. Similarly, a smartphone camera also has a lens to focus light onto a sensor (historically a film!) where the image is converted to a digital signal and is then often processed heavily (as is particularly common in smartphones) before being stored as an encoded format. This makes smartphone cameras more similar to human vision than traditional cameras as the resulting image is much more a product of interpretation and processing in the brain than just a mere sensor reading.

Sound is a physical phenomenon caused by vibration of material and mostly perceived through air as a medium in everyday life. These vibrations propagate through the air as pressure waves. These pressure waves may enter a human ear where these changes in pressure move hair-like fibres in the inner ear, specifically the cochlea. These send electro-chemical impulses to the brain where it is again interpreted as sound.

4 Sound and light waves

Answer the following questions:

a. Explain what sound is and how sound is made?
   A: As explained in the previous exercise, sound is caused by vibration of material and the resulting pressure waves pass into the ear and to the cochlea, an organ in the inner ear with a membrane. Within it, different hair cells are moved by the pressure wave, trigger a neural impulse and this information is passed on to the brain which interprets it as sound.
b. What is the propagation speed of sound?
   i. in the air,

- **A:** 343 m/s (1235 km/h) at 20 °C (https://en.wikipedia.org/wiki/Speed_of_sound)

   ii. in water,

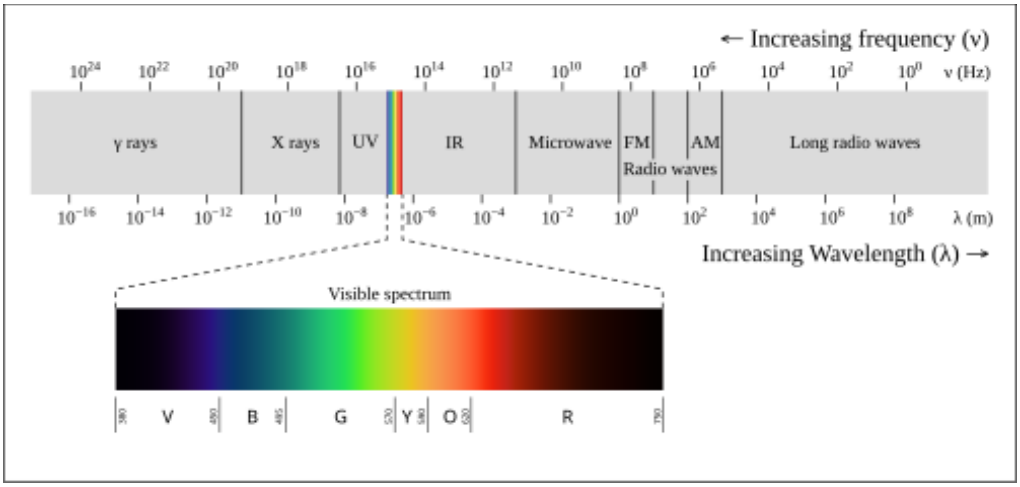
- **A:** 1481 m/s at 20 °C (https://en.wikipedia.org/wiki/Speed_of_sound)

iii. in iron,
A: 5130 m/s (http://hyperphysics.phy-astr.gsu.edu/hbase/Tables/Soundv.html)

iv. in vacuum
A: Since vacuum describes the absence of particles that a pressure wave could propagate through, sound cannot propagate in vacuum.

v. What can you conclude from this?
A: The higher the density of a material is, the faster sound travels through it.

c. What is light? Sketch the electromagnetic spectrum.
A: Light is a relatively small part of the electromagnetic spectrum characterised by human's ability to see it, typical defined as wavelengths between 400 and 700nm. I'll refrain from sketches on this computer as more educational value can be gained from existing illustrations and I have already experienced the educational merits of doing so myself in high school ;).



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d. What is the speed of light?

i. in the air,
A: Refractive index: 1.000293 -> c = 299 704 644,539150029 m/s (Refractive index describes the ration of the speed of light in vacuum to the speed of light in a given optical medium) (https://en.wikipedia.org/wiki/Refractive_index)

ii. in water,
A: Refractive index: 1.333 -> c = 224 900 568.6421605401 m/s (https://en.wikipedia.org/wiki/Refractive_index)

iii. in vacuum
A: 299,792,458 m/s (https://en.wikipedia.org/wiki/Speed_of_light)

e. Let three sound sources (a, b, c) be given. These cause a sound pressure level of a) 0 dB, b) 10 dB, and c) 30 dB. Calculate the sum of the sound pressure levels.
A: As per the formula given in Prof. Schöffmann's slides:

$L = 10 \log_{10}(10^{L1/10} + 10^{L2/10} + \ldots + 10^{Ln/10})$
Here: $L = 10 \log_{10}(10^{0/10} + 10^{10/10} + 10^{20/10}) = 10 \log_{10}(10^0 + 10^1 + 10^2) = 10 \log_{10}(111) = 20.453229\ldots \approx 20.45$ dB

5 Sound parameters

Explain the physical parameters frequency, amplitude and phase of a wave. Explain how these parameters are related to the properties of loudness (i.e., sound level), pitch, and timbre of a sound wave.

A: When given a wave, its peaks (or troughs) have a certain distance from each other. This distance is referred to as wavelength. Frequency describes how many of such cycles pass a point per second. Given that a wave moves at a certain known speed, the frequency and wavelength are directly related; if the wavelength is longer, more time will be needed to complete a cycle and thus the frequency is lower and vice versa. Generally, the wavelength can be expressed as wavelength = speed of wave / frequency.

Amplitude describes how high or intense the peak of a wave is.

Phase describes the offset of a wave to another one. Phase typically uses angle-like quantities to represent this, where 360° or 2*Pi radians mean that a full cycle has been completed and two waves are “in phase” again.

The pitch of a sound is determined by the frequency/wavelength that the wave has, higher frequency results in a higher perceived pitch and lower frequency in lower pitch.

The loudness of a sound corresponds to the amplitude of its associated wave; higher amplitude results in a louder sound

Timbre is a more complex concept. It is perceived when e.g. two instruments or two singers perform a note at the same pitch, however they clearly sound distinct. This is because not only a single frequency is present in such a sound but is rather a mix of several frequencies in addition to the fundamental frequency. When multiples of the fundamental frequency are present, we perceive the resulting sound as a musical tone. However, when predominantly unrelated frequencies are present, it appears as unmusical noise.

Sources

All exercises were completed using the latest lecture slides from Klaus Schöffmann for the course “Introduction to Multimedia” (winter semester 2026) without any use of large language models.

All additional complementary sources are listed below:

- https://de.wikipedia.org/wiki/Hypermedia

- https://my.clevelandclinic.org/health/articles/21204-vision
- https://en.wikipedia.org/wiki/Speed_of_sound
- http://hyperphysics.phy-astr.gsu.edu/hbase/Tables/Soundv.html
- https://en.wikipedia.org/wiki/Speed_of_light
- https://en.wikipedia.org/wiki/Refractive_index